

Your Name and ID:

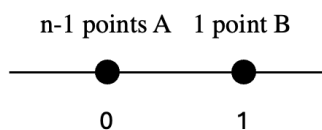
Your class: ☐ A (Prof. A. Pietracaprina) ☐ B (Prof. F. Silvestri)

REMINDERS:

- Write your Name and ID on every sheet that you return
- Excessively long answers will be penalized
- TOTAL TIME: 2 hours

PART 1

1. [3 points] With reference to the Resilient Distributed Dataset (RDD) of Spark, define the notion of *lazy evaluation* and explain how it may negatively affect the quality of time measurements.
2. [3 points] Consider a set P of N points from a metric space, and a subset $T \subseteq P$ of k centers such that for every $x \in P$, $d(x, T) \leq R$. Let Δ and Δ_T denote the diameters of P and T , respectively. State and prove an upper to Δ as a function of Δ_T and R .
3. [3 points] Suppose that Sticky Sampling is used to determine frequent items in a stream Σ of n items, with frequency threshold φ and parameters ε and δ . Let S be the set of items returned by the algorithm. What are the probabilistic guarantees regarding S ?
4. [4 points] (QUESTION RELATED TO THE HOMEWORKS) Consider a set of points $U \subset \mathbb{R}^d$, split into two demographic groups A and B , that is, $U = A \cup B$. Given a number of clusters k , fair k -means clustering aims at finding a set C of centroids which minimize the fair objective function $\Phi(A, B, C)$.
 - (a) Define $\Phi(A, B, C)$.
 - (b) Suppose that 1 centroid must be chosen for the set of n points in $\mathbb{R}$ depicted in the image below. If $n \rightarrow +\infty$, what is a good centroid for the standard k -means objective and what is a good centroid for the above fair objective?



PART 2

1. **[7 points]** Let G be a $t \times t$ grid of *cells* covering a region of a mobile network, and let P be a dataset of N *connections* to the cells of G . Each connection is represented by a pair $(k, (i, j))$ where $k \in [0, N - 1]$ is a distinct integer key, and (i, j) , with $0 \leq i, j < t$, are the row and column indices of the cell used by the connection. Assume $t = O(\sqrt{N})$.
 - (a) Design a **3-round** MapReduce algorithm that returns the rows of G such that more than $t/2$ of their cells have been used by connections of P . For each such row i , the algorithm must return a pair (i, t_i) , where $t_i > t/2$ is the number of distinct cells of row i were used by connections of P . No assumptions must be made on the number of connections per cell.
 - (b) Analyze the local and aggregate space, M_L and M_A , of the algorithm. For full score, you must get $M_L = o(N)$ and $M_A = O(N)$.
2. **[6 points]** Let P be a set of n D -dimensional Boolean vectors. Suppose that P is stored into a hash table T built using a hash function h randomly drawn from the bit-sampling LSH family $\mathcal{H} = \{h_i : 0 \leq i < D\}$. For $h \in \mathcal{H}$, let "**not**($h(x)$)" denote the negation of the binary value $h(x)$.
 - (a) Given two vectors p and q , determine the probability $\Pr_{h \in \mathcal{H}}[h(p) = \text{not}(h(q))]$ as a function of the Hamming distance $d_H(p, q)$.
 - (b) Given a query vector q , we want to find an r -far vector $p \in P$, i.e., such that $d_H(p, q) \geq r$. Based on the above analysis, how would you efficiently search such a p in the table T ? What probabilistic guarantees does your method provide?